

CSF 250

Electricity: static, current

Electrical quantity: Charge (in Coulomb, C), $e = 1.6 \times 10^{-19} \text{ C}$ (smallest charge than can exist by itself)
Charges are conserved (i.e. neither created or destroyed)

Electric potential energy: e.g. when you try to bring two similarly charged objects together, the work done against the repulsive force will be converted to electric potential energy. If you release them, the E.P.E will be converted to K.E as they move apart again (repelled).
(Energy conserved)

Instead of K.E we can use this E.P.E to convert into useful energy like light energy in bulbs.

G.P.E: need energy to go up as gravity pulls u down. $U = mgh$

Potential energy - depends on reference point. (0 at ref point)

Work done/Energy converted - doesn't depend on reference point -

Gravitational potential: $V = gh$ (also depends on ref point) in units of J/kg

$\therefore$ G.P.E, $U = mV$, $W = m\Delta V$

But g might not always be 9.8 . So, $U = mGM \left(\frac{1}{R} - \frac{1}{R+h} \right)$

Gravity is always attractive, unlike electric force.

Electric potential, Voltage: V (scalar)

Electric potential energy, $U = qV$ or $W = q\Delta V$

- Object's properties - energy, m , q , v (sth that the object possesses)
- Environment effect - Work, potential, F (sth that is inflicted upon an object)
- Everything likes to go from high to lower potential energy. ($\Delta W = +ve$)
- It requires energy to go from low to higher P.E. ($W = -ve$)

$+ve$ charges tends to go from high to low P.E. and high to low Potential
 $-ve$ charges tends to go from high to low P.E. and low to high Potential

Still require energy to go the other way.

ex.

$q = -3C$

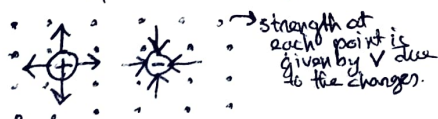
$V_1 = -6V$
 $V_2 = 18V$

$V_1 = -1V$
 $V_2 = 3V$

The charge would go left to right (low to high potential)

$\Delta V = -6 - (-1) = -5V$

$W = q\Delta V = (-3) \times (-5) = 15J$



Current: moving charges. I (in Amps). $I = \frac{q}{t}$ in C in sec

Extra: Using Desmos graphing Calculator.

You can plot graphs with constant or changing variables; points that can be set as draggable or not by right clicking the circle; tables by clicking '+' at top left and then table. You can also add files and images using the '+'. You can also make the points in g table draggable.

There's a simulator in Week 2 after Lec 5. (very cool)

②

Voltage difference causes currents to flow. e.g. battery in a circuit.

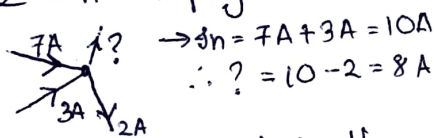
Current is positive in the direction of a moving positive charge.

3A of positive ion moving left = -3A of +ve ion moving right = 3A of -ve moving right.

So, we can't tell if +ve or -ve charge is moving.

∴ Voltage diff causes current to flow from high to lower V, regardless of charge type.

Current is scalar. The direction is from current density (vector).
So we can simply add the currents, instead of using the Δ rule.



$\vec{c} = \vec{a} + \vec{b}$
 $c^2 = a^2 + b^2$

Direction of current is the direction of movement of +ve charges.
The charges are accelerating because of electric force. Also because its Potential energy decrease, increasing K.E as it moves away.

Insulators: e^- are bound to atoms and not free to move.
No current will flow

Conductors: When V is applied, the e^- move from atom to atom from high p.d to low p.d. As they bump into each atoms, they lose their speed unlike in vacuum, and accelerate again and so on.
So, e^- 's will have average and maximum speed in conductors. Hence, constant current. (if R same)

Speed increases forever

Ohm's Law: $I = G \Delta V$ conductance, Ω^{-1} , $G_{copper} > G_{wood} \rightarrow \approx 0$

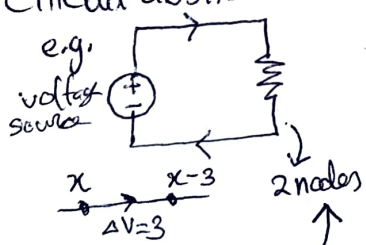
Resistance, Ω
 $R = \frac{1}{G}$
 $V = RI$

Resistance is the property of the conductor. Stays almost const for that material conductor

Power = $IV = \frac{qV}{t}$ energy lost per time

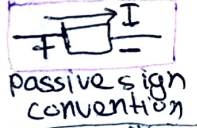
Energy lost = qV p.d

Circuit abstraction:

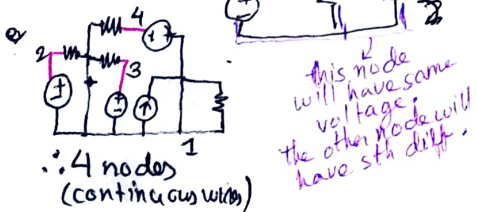
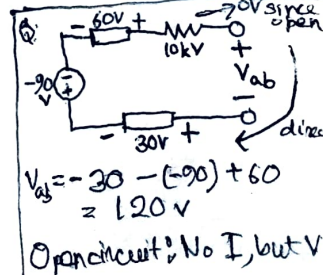
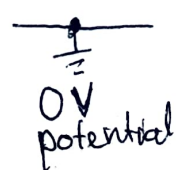
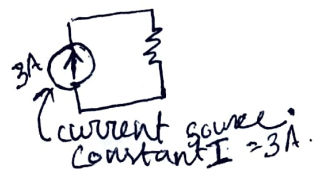


Golden Rules:

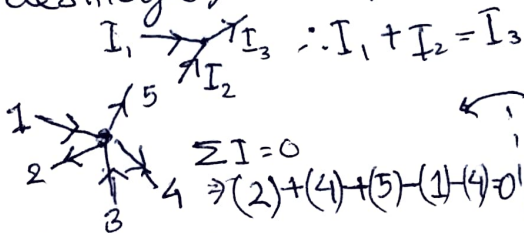
- 1. A continuous piece of wire, will have the same voltage everywhere. (1 node for each voltage). $\Delta V = (V_+) - (V_-)$
- 2. 1 component for each current.



Power consumed: +ve
Power supplied: -ve



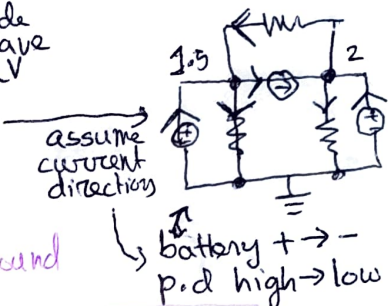
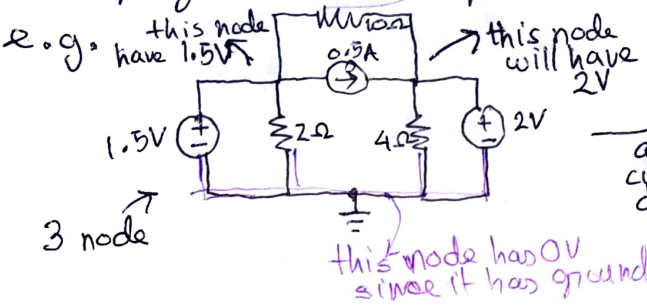
• Kirchhoff's current Law (KCL): since current can't be destroyed, sum of I entering a point = sum of I leaving
 On, Total out - Total in = 0
 On, $\sum I = 0$



if current is exiting a node $\rightarrow +ve$
 " " "entering" " $\rightarrow -ve$

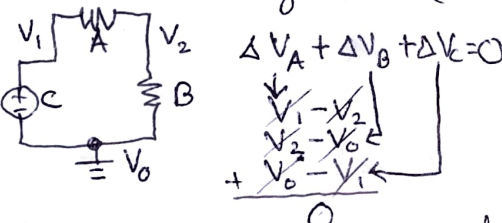
The Algebraic Sum of all Currents in a Node is Zero.

$P = I^2 R$
 $= \frac{\Delta V^2}{R}$



Write KCL equation for each node.
 Use $V=RI$

• Kirchhoff's Voltage law (KVL): $\sum \Delta V = 0$



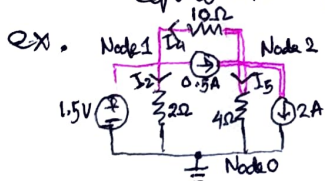
voltage diff across each component.

The Algebraic Sum of all Voltage difference in a Loop is Zero.

• Node Analysis: Calculate all the node voltages using KCL. Using that, we can calculate anything. Step:

1. Identify all the nodes and mark them from 0 to N. $\rightarrow$ ground node, $V=0$
2. Write equations for all voltage sources $\rightarrow \Delta V = \text{labeled value}$
3. Write the currents through the resistors using Ohm's law. (Must follow positive sign convention)
4. Write KCL/Node Equations for all the nodes and supernodes (especially for the ones with unknown V)
5. Get all node voltages by solving all the equation. (use calculator to directly solve linear equation)

always minus



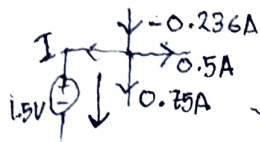
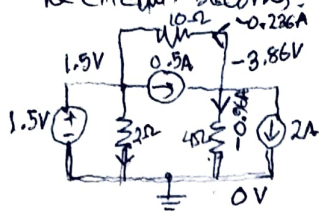
Node 0: $V_0 = 0$ (ground) always
 Eqn for voltage source: $V_1 - 0 = 1.5V$
 $\Rightarrow V_1 = 1.5$

pick a current direction Eqn for 10Ω resistor: $I_4 = \frac{V_2 - V_1}{10} = \frac{V_2 - 1.5}{10}$
 " " 2Ω " : $I_2 = \frac{V_1 - 0}{2} = \frac{1.5}{2}$
 " " 4Ω " : $I_5 = \frac{V_2 - 0}{4}$
 KCL equation for node 2: $I_4 + I_5 + 2 - 0.5 = 0$
 Substitute and solve to get V_2

not node 1 since we already know V_1

(4)

The circuit becomes:



I is to - in voltage source

positive power = consuming power from circuit.

ex. ~~10Ω~~ resistor = $\frac{V^2}{R} = 1^2 R$

negative power = supplies power.

ex. current source: $P = \Delta V I$

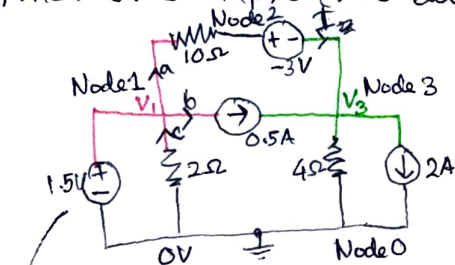
$$= (V_2 - 0) I = (-3.86 - 0) \times 2 = -7.72 \text{ W}$$

Power of voltage source: $-V_2$

$$I = 0.236 + 0.5 + 0.75 = 1.486$$

$$\therefore P = 1.5 \times (-1.486) = -2.23 \text{ W}$$

Another example (two batteries)



$$\text{Current} = a + b - c = \frac{V_1 - V_2}{10} + 0.5 - \frac{V_2 - V_1}{4}$$

Node 1 Battery: $V_1 - 0 = 1.5$
 $V_1 = 1.5 \text{ V}$ [Node 1]

right battery: $V_2 - V_3 = -3$ [Super node: Node 2, 3] (i)

Node 3: $\frac{V_3}{4} - \frac{0}{4} + 2 - 0.5 - I = 0$ (ii)
 (all the currents in and out of that node)

Node 2: $\frac{V_2}{10} - \frac{V_1}{10} + I = 0$ (iii)

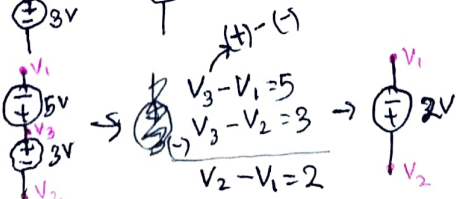
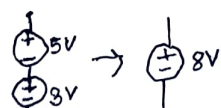
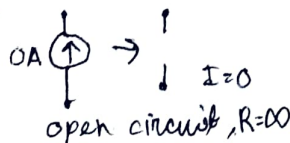
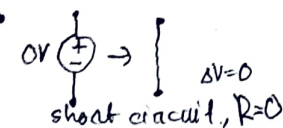
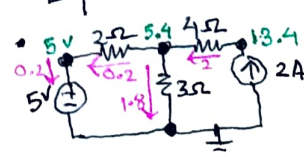
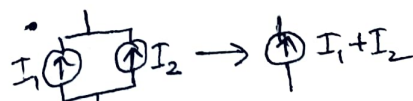
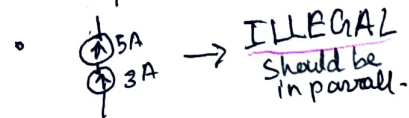
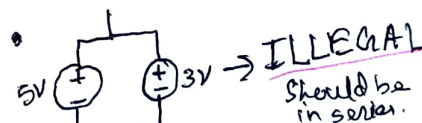
Combine all three equations to get V_2 and V_3
 (I gets cancelled out with (ii) + (iii))

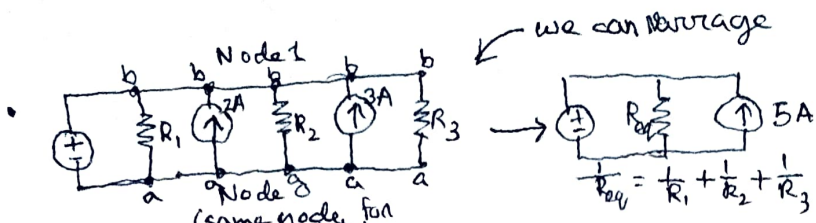
Node n eqn: $\frac{V_n - V_1}{R_1} + \frac{V_n - V_2}{R_2} + \frac{V_n - V_3}{R_3} = 0$

Equivalent circuit:

• $V_1 - V_3 = IR_1$
 $V_3 - V_2 = IR_2$
 $V_1 - V_2 = I(R_1 + R_2)$
 • Should have: the same overall circuit equation; the same p.d across its edge; same I

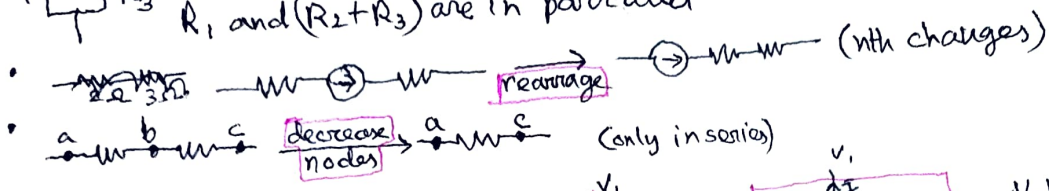
• $4 \frac{1}{2} \text{ } \rightarrow 12 \text{ } \rightarrow 2$
 end nodes similar (same ΔV)
 $I_1 = \frac{1}{4} (V_2 - V_1)$
 $I_2 = \frac{1}{6} (V_2 - V_1)$
 $I_3 = \frac{1}{12} (V_2 - V_1)$
 $I_1 + I_2 + I_3 = I = \frac{1}{2} (V_2 - V_1)$
 same



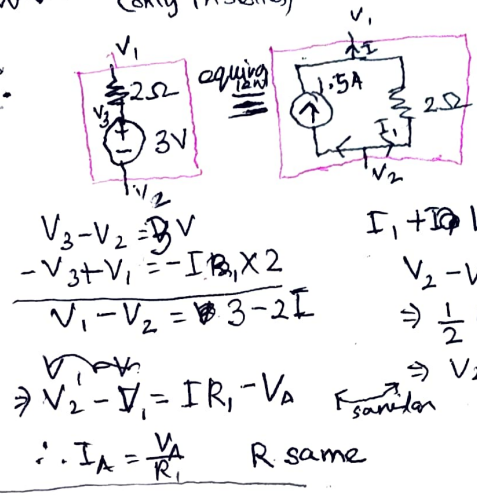


Parallel: Same node
Series: Same I

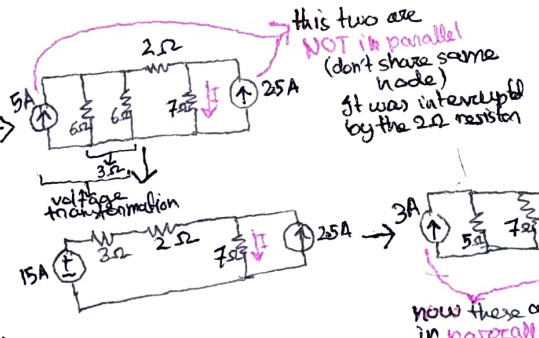
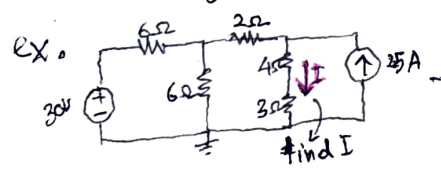
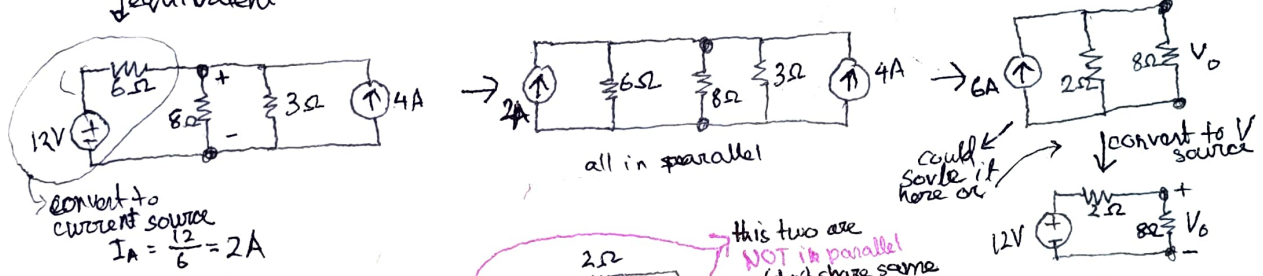
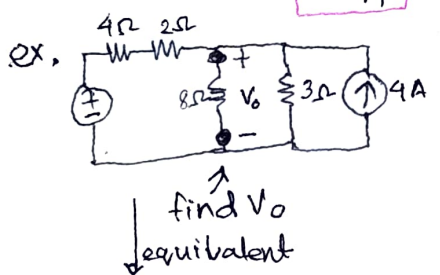
R_1 and R_2 are not in parallel [Not the same node]
 R_1 and $(R_2 + R_3)$ are in parallel



Source transformation:



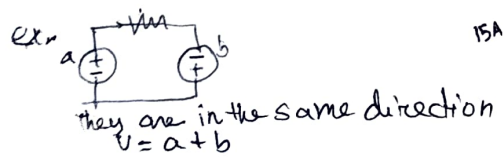
Voltage source $\rightarrow$ Current source



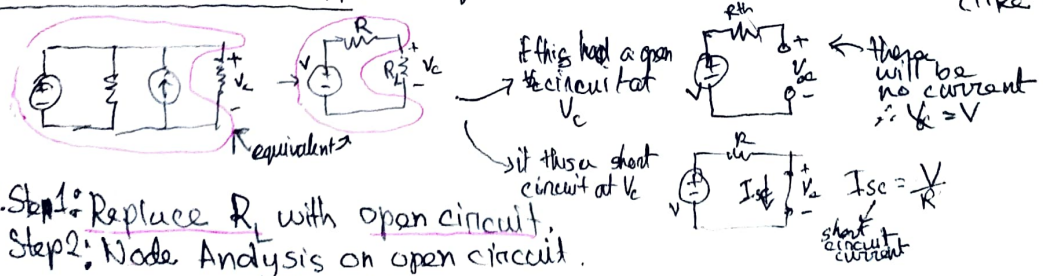
$V_0 = \frac{8}{8+2} \times 12 = 9.6V$

$\therefore I = 5.5 \times \frac{5}{7+5} = 2.29A$

in the 7 ohm resistor since 7 ohm will have less current.



Thevenin's Theorem: simplifies a circuit by find equivalent circuit. (like prev examples)

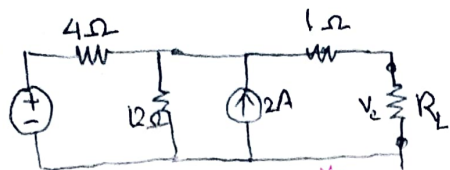


- Step 1: Replace R_L with open circuit.
- Step 2: Node Analysis on open circuit.
 $V_{oc} = V$
- Step 3: Replace R_L with short circuit.
- Step 4: Solve again to get I_{sc} . $R = \frac{V_{oc}}{I_{sc}}$

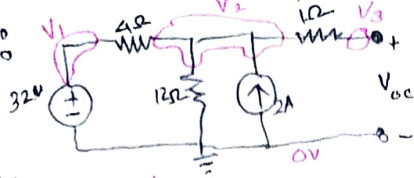
Now u can put any thing in that V_c place and solve easily

⑥

ex.



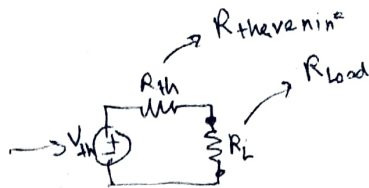
Step 1:



Step 3:

Node Analysis to get V_1, V_2

$$I_{se} = \frac{V_2 - 0}{1\Omega} = 7.5A, \quad R = \frac{V_{oc}}{I_{se}} = \frac{30}{7.5} = 4\Omega$$



Node analysis to get V_1, V_2 [On use Equivalent transformation]

$$\begin{aligned} V_1 &= 32 \\ V_2 \left(\frac{1}{4} + \frac{1}{12} + \frac{1}{1} \right) - \frac{32}{4} - \frac{V_1}{1} - 2 &= 0 \\ V_1 &= V_3 \text{ (Cos no } \downarrow \text{ through that resistor)} \\ V_2 \left(\frac{1}{4} + \frac{1}{12} + \frac{1}{1} \right) - \frac{32}{4} - \frac{V_2}{1} - 2 &= 0 \\ \text{Get all the node voltages} \\ V_{oc} &= V_3 - 0 \rightarrow 30 \\ V &= V_c = 30 \end{aligned}$$

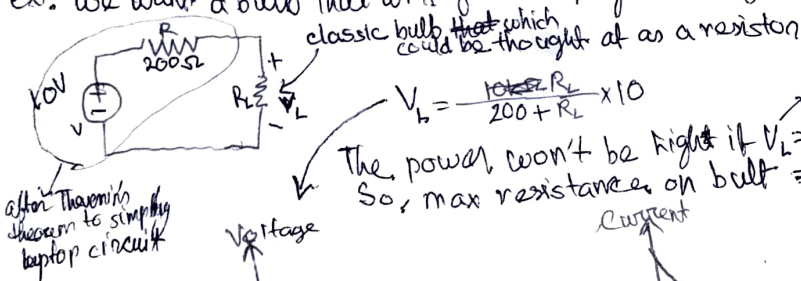


[Another ex. could be sockets $\left[\begin{smallmatrix} \circ \\ \circ \end{smallmatrix} \right]$. They are open circuits with a complicated circuit behind it. But can use a multimeter to get the Voltage]

• Thevenin's theorem works on Linear circuits only.

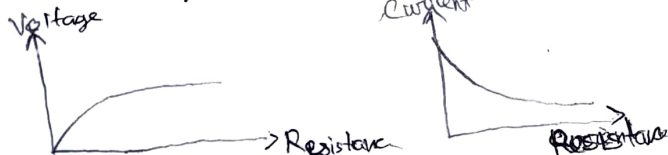
• Maximum Power Transfer:

ex. we want a bulb that will glow the brightest in your laptop:



$$V_b = \frac{10 \times R_L}{200 + R_L} \times 10$$

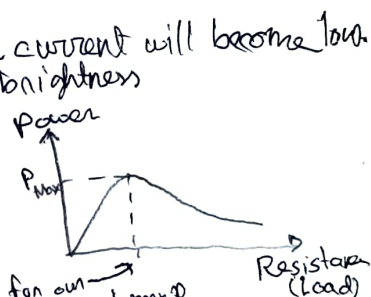
The power won't be right if $V_b = 10$ since current will become low. So, max resistance on bulb $\neq$ max brightness



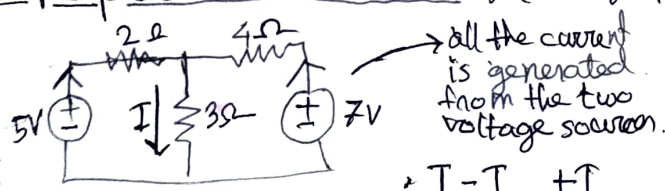
$\therefore$ We get brightest when $R_L = R$ (max power)

From the Thevenin

max voltage. $P = VI$



• Superposition Theorem (adding up currents)



$$I = I_{5V} + I_{7V}$$

$\therefore$ We add all the current from sources.

This simplifies the circuit and makes calculations easier.

Step 1: keep 1 independent source and kill all other.

Step 2: Circuit analysis to solve current/voltage component of that source.

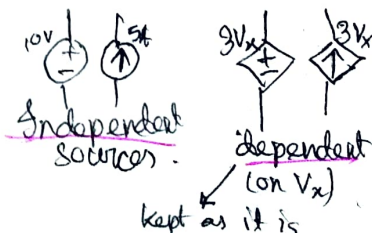
Step 3: Repeat for all independent sources

Step 4: Add them up.

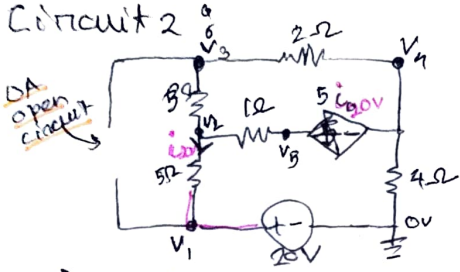
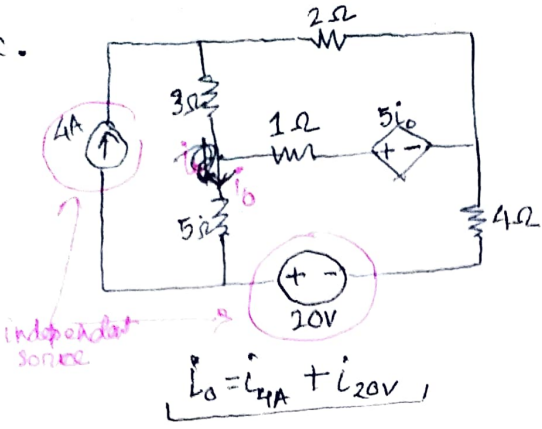
we now find each of the source currents separately. for I_{5V} , we will only keep the 5V source:



Do node analysis on sth to get I_{5V} . Do the same to get I_{7V}

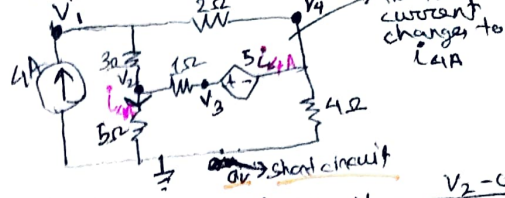


ex.



Do node analysis again and get
 $i_{20V} = \frac{V_2 - V_1}{5} = -3.53A$

Circuit 1:
 (Step 1) For 4A source!

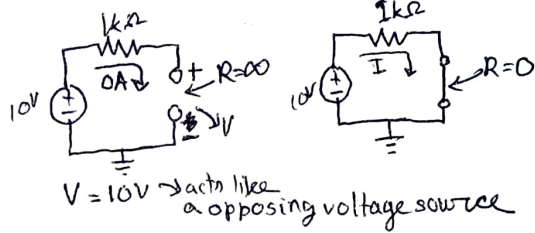
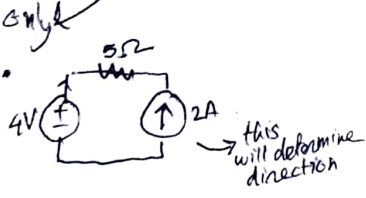


Node analysis: $V_3 - V_4 = \frac{V_2 - 0}{5\Omega} \times 5$
 $\Rightarrow V_3 - V_4 - V_2 = 0$ — (1)
 $V_1(\frac{1}{3} + \frac{1}{2}) - \frac{V_2}{3} - \frac{V_4}{2} - 4 = 0$ — (Node 1)
 $V_2(\frac{1}{1} + \frac{1}{3} + \frac{1}{5}) - \frac{V_3}{1} - \frac{V_4}{3} = 0$ — (Node 2)
 Supernode-3, 4:
 $V_3(\frac{1}{1}) - \frac{V_2}{1} + V_4(\frac{1}{2} + \frac{1}{4}) - \frac{V_1}{2} = 0$
 Node 4
 Solve to get V_2 and then $i_{4A} = \frac{V_2 - 0}{5} = 3.06A$

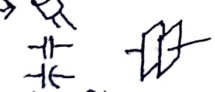
Step 4: Add

$i_0 = i_{4A} + i_{20V}$
 $= -0.47A$

Why does superposition theorem work? Current and voltage can superpose (but not power). We can add them up. Because linear circuit (all node equations are linear). All the variable are single at a by t...

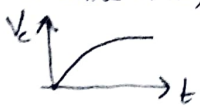


Capacitors & Inductors



store Charges gather in one sheet. Store Electrical energy.

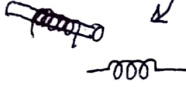
Capacitor is like a open circuit, acts like opposing battery after some time)



$Q \propto V$ — capacitance (Farad, F)
 $\Rightarrow Q = CV$

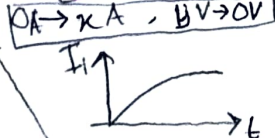
$\Rightarrow \frac{dQ}{dt} = C \frac{dV}{dt}$

$\Rightarrow I = C \frac{dV}{dt}$



Coil wire creates changing magnetic field. Stores Magnetic energy.

Inductor turns into a straight wire after some time. (Short circuit)



$0A \rightarrow xA, yA \rightarrow 0A$

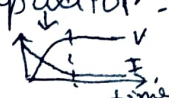
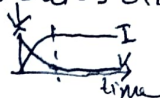
Capacitor reacts to change in voltage

Henry, H

$V = L \frac{dI}{dt}$

Inductor reacts to change in current.

- Capacitor
-
- 10V
- 1k Ω
- $V_R(t)$
- $t=0$
- i.e. we are turning it On at $t=0$
- $V(t)$
- $i(t)$
- 1 μ F

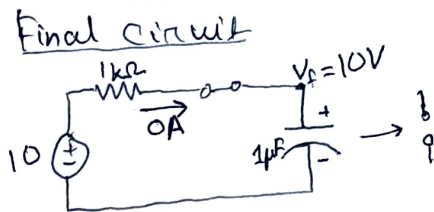
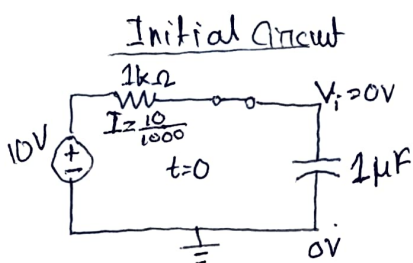


$$V(t) = V_f + (V_i - V_f) e^{-\frac{t}{\tau}}$$

$$\tau = RC$$

→ time constant

$$\tau = \frac{L}{C} \rightarrow \text{for inductor}$$

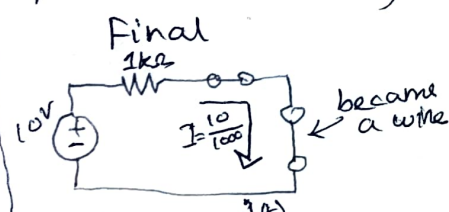
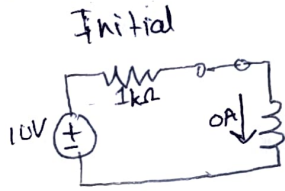
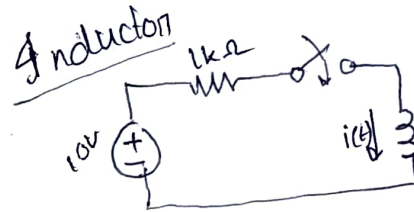
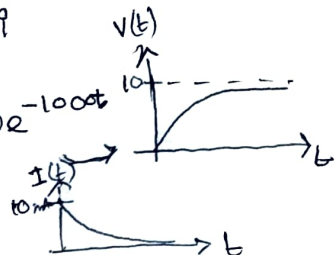


$$i(t) = i_f + (i_i - i_f) e^{-\frac{t}{\tau}}$$

$$\therefore v(t) = 10 + (0 - 10)e^{-\frac{t}{10^{-3}}} = 10 - 10e^{-1000t}$$

$$\therefore i(t) = 0 + (10 - 0)e^{\frac{-t}{10^{-3}}} = 10e^{-1000t}$$

(Changing)



$$i(t) = 10 + (0 - 10)e^{-\frac{t}{0.005}}$$

sin mA

- ~~Charging~~ Discharging:



$$Z = 5 \times 1 = 5$$

$$V_c = 5 \times 1 = 5$$

$$V_{c_j} = 10 \text{ V} , V_{c_f} = 0 \text{ V} , V_c(t) = 10 e^{-\frac{t}{5}}$$

$$i_{c_f} = -2A, \quad i_{c_f} = 0V, \quad i_c(t) = -2e^{-\frac{t}{5}}$$

Thomas Edison
(light bulbs)

right after opening switch
when $\frac{1}{s} \approx \oplus$. So current up.

- DC = Direct current $\rightarrow$ HDD, SSD

Transient $\equiv$ (something that only lives for a short time)

- AC = Alternating current → lights, High power, electrical components

Bd uses 50 Hz a.c. and 220V.

↳ so current doesn't have to move along the wire

That oscillates back & forth

So less energy lost.

(More efficient than DC)

(Good for transmission)

Using transformers)

Transformers works in ac only.

if you use 1.5
100V and 60Hz charger
in Bd, it might
blow up;

magnetic field (C) $\uparrow$ II
wire

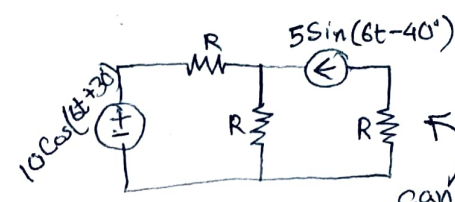


 a.c
 changing mag field
 Electric $\longleftrightarrow$ Magnetic

$I_p V_p \approx I_s V_s$
 high small

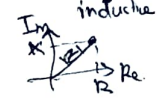
• AC voltages:

since it's oscillating, we use sin or cos



can have capacitors and inductors in a similar way. They are called Impedance, Z . We will use complex no. to solve.

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

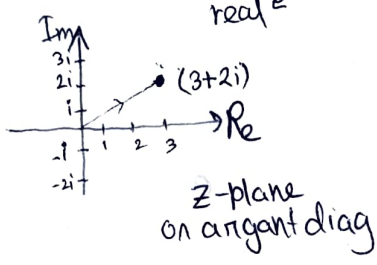


• Complex no. Revision:

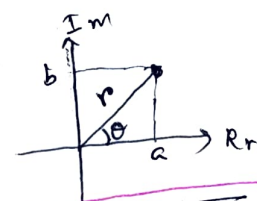
$$i = \sqrt{-1}$$

Complex no. $a + bi$
 a → real
 bi → imaginary

$$\mathbb{R} \subseteq \mathbb{C}$$



• Addition



$$\begin{aligned} r &= |z| = \sqrt{z \cdot \bar{z}} \\ a &= r \cos \theta, \quad b = r \sin \theta \\ z &= a + bi = r (\cos \theta + i \sin \theta) \\ &= r e^{i\theta} \end{aligned}$$

↙ cartesian / rectangular

↘ Polar

• Adding two $\mathbb{C}$ no.:

$$\begin{aligned} z_1 + z_2 &= (3+2i) + (-5+5i) \\ &= -2+7i \end{aligned}$$

• Multiply:

$$\begin{aligned} (3+2i)(-5+5i) &= -15 + 15i - 10i + 10i^2 \\ &= -25 + 5i \end{aligned}$$

• Multiply: $(3e^{i\frac{3\pi}{4}})(2e^{i\pi})$

$$= (3 \times 2) e^{i(\frac{3\pi}{4} + \pi)}$$

• Divide:

$$\begin{aligned} \frac{3+2i}{-5+5i} &\rightarrow \text{conjugate} \\ &= \frac{3+2i}{-5+5i} \times \frac{-5-5i}{-5-5i} \\ &= \frac{(3+2i)(-5-5i)}{(-5)^2 - (5)^2(i)^2} \\ &= \frac{-5-25i}{50} \end{aligned}$$

$$\begin{aligned} \text{Divide} &= a e^{im} \div b e^{in} \\ &= \frac{a}{b} e^{i(m-n)} \end{aligned}$$

In electricity we'll use $j = \sqrt{-1}$ also in python

and $r e^{i\theta} \rightarrow r \angle \theta$

$$\text{e.g. } (5 \angle 30)(2 \angle 10) = 10 \angle (30+10) = 10 \angle 40$$

If you multiply $1 \angle \theta$ to a complex no. z , it rotates it counterclockwise by θ . e.g. $(3 \angle 30) \times (1 \angle 10) = (4 \angle 40)$

$$\begin{aligned} \sqrt[3]{1} &= 1, \frac{-1+\sqrt{3}i}{2}, \frac{-1-\sqrt{3}i}{2} \\ &= 1 \angle 120^\circ, 1 \angle -120^\circ \end{aligned}$$



$$\begin{aligned} 1^3 &= (1 \angle 120)^3 \\ &= 1 \angle (120 \times 3) \\ &= 1 \angle 360 = 1 \angle 0 = 1 \end{aligned}$$

$\sqrt[5]{1} \rightarrow$ divide 360 in 5 parts.

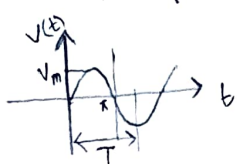


$$\therefore 1, 1 \angle 72^\circ, 1 \angle 144^\circ, 1 \angle -144^\circ, 1 \angle -72^\circ \text{ (easier to calculate)}$$

$r \angle \theta \rightarrow$ phasors

10

- DC - time invariant $\begin{matrix} v \\ \rightarrow \\ t \end{matrix}$
- AC - time variant $\begin{matrix} v \\ \rightarrow \\ t \end{matrix}$ $\rightarrow$ sinusoids - sin or cos functions



$$V(t) = V_m \sin(\omega t + \phi)$$

amplitude

angular frequency
 $f = \frac{2\pi}{\omega}$

phase, argument (where the signal starts)

$$V_m \angle \phi$$

phasor

- Impedance - same as resistance but in complex numbers.

e.g. $Z \rightarrow 5 + j6 \Omega$

Impedance of an inductor: $\frac{L=10H}{\omega=4} \rightarrow Z_L = j\omega L = j \times 4 \times 10 = j40$

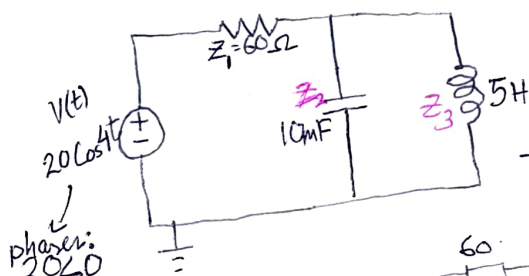
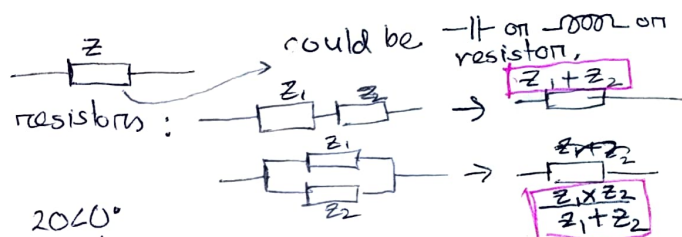
" " capacitor: $\frac{C=10\mu F}{\omega=4} \rightarrow Z_C = \frac{1}{j\omega C} = \frac{1}{j \times 4 \times 10^{-2}} = \frac{1}{j25} = -j25 \Omega$

" " resistor: $Z_R = 60 \Omega = R$

$$V = IZ$$

- Impedance Combinations:

It follows the same rules as resistors:

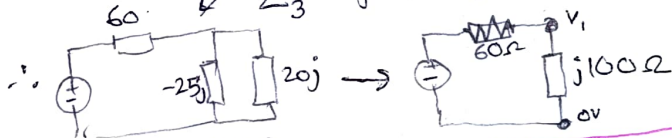


$\omega = 4$

$Z_1 = 60 \Omega$

$Z_2 = \frac{1}{j\omega C} = \frac{1}{j \times 4 \times 10^{-2}} = -j25 \Omega$

$Z_3 = j\omega L = j20 \Omega$



$Z_p = \frac{Z_2 \times Z_3}{Z_2 + Z_3}$

$= \frac{-j25 \times j20}{-j25 + j20} = \frac{500}{-j5} = j100 \Omega$

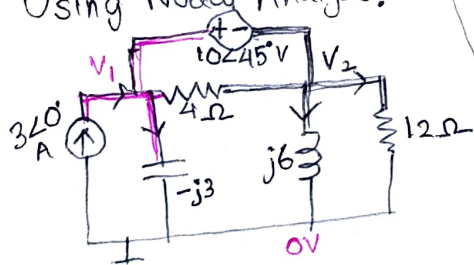
$V_1 = \frac{Z_p}{Z_1 + Z_p} \times V(t)$

$= \frac{j100}{60 + j100} \times (20 \angle 0^\circ)$
 $= (0.735 + j0.44)(20 \angle 0^\circ)$
 $= (0.857 \angle 30^\circ)(20 \angle 0^\circ)$
 $= 17.41 \angle 31^\circ$

Voltage across the capacitor and inductor.

AC circuit combination

Using Nodal Analysis:



Supernode at V where we don't know either voltage. $V_1 = V_2 + 10 \angle 45^\circ$

Node 1: $-3 \angle 0 + \frac{V_1 - 0}{-j3} + \frac{V_2 - 0}{j6} + \frac{V_2 - 0}{12} = 0$

& 2: $3 \angle 36 - \frac{1}{4}(4V_2 + 10 \angle 45) + \frac{1}{j} 2V_2 + V_2 = 0$

$\rightarrow 3 \angle 36 - \frac{1}{4}(4V_2 + 40 \angle 45) + j2V_2 + V_2 = 0$

multiply by 12

$\rightarrow -36 + j4V_2 + j40 \angle 45 - jV_2 + V_2 = 0$
 $\rightarrow 40 \angle 135$

$\frac{1}{j} = -j$
 $j = 1 \angle 90$

$$\Rightarrow -36 + j4V_2 + 40\angle 135 - j2V_2 + V_2 = 0$$

$$\Rightarrow V_2(j2+1) = 36 - 40\angle 135$$

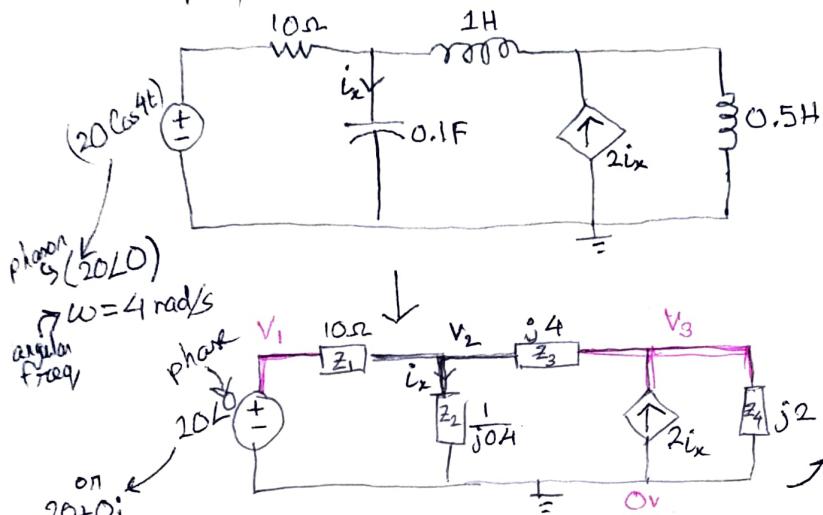
$$\Rightarrow V_2 = \frac{36 - 40\angle 135}{1+j2} = 31.4\angle -87.18$$

$$\therefore V_1 = (31.4\angle -87.18) + (10\angle 45) = 25.78\angle -70.48$$

use calculator

in calculator:
Go to 'complex mode'
Type the equation exactly.
Use ' $\angle$ ' and 'E' for ' e '.
Press '=' for the complex ans.
For polar ans:
[OPTN] $\rightarrow$ [V] $\rightarrow$ [I] $\rightarrow$ 'r $\angle$ '
[=]

Example:



Step 1: $\omega = 4$

Step 2: Convert voltage/current source

e.g.

$$\begin{aligned} 3\sin(4t+30^\circ) &= 3\cos(4t+30-90) \\ &= 3\angle 60 \end{aligned}$$

Step 3: Convert R, capacitor, inductor to Z
Step 4: Do Node-analysis and solve the complex linear equations.

Nodal analysis:

$$V_1 = 20\angle 0$$

Node 2:

$$V_2\left(\frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3}\right) - \frac{V_1}{Z_1} - \frac{0}{Z_2} - \frac{V_3}{Z_3} = 0$$

$$i_x = \frac{V_2 - 0}{Z_2}$$

Node 3:

$$V_3\left(\frac{1}{Z_3} + \frac{1}{Z_4}\right) - \frac{V_2}{Z_3} - \frac{0}{Z_4} - 2i_x = 0$$

$$V_2\left(\frac{1}{10} + \frac{1}{j0.4} + \frac{1}{j4}\right) - \frac{V_1}{10} - \frac{V_3}{j2} = 0$$

put this in calculator to get the complex no.

Get the 3 equations:

$$V_1 = 20$$

$$(-0.1)V_1 + (0.1 + j0.15)V_2 + (j0.25)V_3 = 0$$

$$0 + (-0.55j)V_2 + (-j0.75)V_3 = 0$$

Put these coefficient in a online complex linear equation calculator and get ans:

$$\begin{pmatrix} 1 & 0 & 0 \\ -0.1 & 0.1 + j0.15 & j0.25 \\ 0 & -0.55j & -0.75j \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} = \begin{pmatrix} 20 \\ 0 \\ 0 \end{pmatrix}$$

* Don't forget to give '*' between $j0.15$
 $\therefore "j * 0.15"$

Ans: $V_1 = 20 = 20\angle 0$

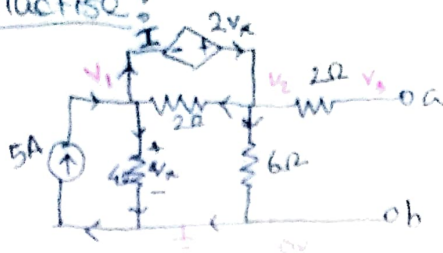
$$V_2 = 18 + j6 = 18.97\angle 18.43$$

$$V_3 = -13.2 - 4.4j = 13.91\angle -161.57$$

$$v_3(t) = 13.91 \cos(4t - 161.57)$$

amplitude ω phase
same method for i_x (in Ques 4)

Practise



$$V_2 - V_1 = 2V_x$$

$$V_x = V_1$$

$$\Rightarrow V_2 = 3V_1$$

$$\text{Node 1: } 5 + \frac{V_2 - V_1}{2} = \frac{V_1 - 0}{4} + I$$

$$\text{Node 2: } I = \frac{V_2 - V_1}{2} + \frac{V_2 - 0}{6} + 0 \quad \rightarrow 2\Omega \text{ resistor has no current (open circuit)}$$

$$5 = \frac{V_1}{4} + \frac{V_2}{6}$$

$$\Rightarrow 5 = \frac{V_1}{4} + \frac{3V_1}{6}$$

$$\Rightarrow V_1 = \frac{20}{3}, \quad V_2 = 20$$

$$\text{Node 3: } \frac{V_2 - V_2}{2} = 0 \quad \leftarrow \text{no current}$$

$$\Rightarrow V_3 = V_2$$

$$\Rightarrow V_3 = 20$$

Step 2: a to b = 20V $\downarrow$

$$V_2 = 3V_1$$

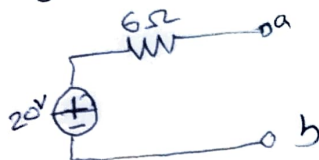
$$\text{Node 1 and 2: } 5 + \frac{V_2 - V_1}{2} = \frac{V_1 - 0}{4} + \frac{V_2 - V_1}{2} + \frac{V_2 - 0}{6} + \frac{V_2 - 0}{2}$$

$$\Rightarrow 5 = \frac{V_2}{3 \times 4} + \frac{2}{3} V_2$$

$$\Rightarrow V_2 = \frac{20}{3}$$

$$I_{ab} = \frac{V_2 - 0}{2} = \frac{\frac{20}{3} - 0}{2} = \frac{10}{3}$$

$$R = \frac{V_2}{I_{ab}} = \frac{20}{\frac{10}{3}} = 6\Omega$$



CSE250 Lab

Multimeter: Red and Black probs (wires). 3x holes/plugs

Rotating Knob.

- V- (for measuring dc voltage)
- V~ (for measuring ac voltage)
- A- (measure dc current. Avoided since have to connect in series on meter damaged)
- Ω
- ⚡ (checks the condition of a diode)

the no. given is the max that it can measure in than range. Set the knob at a no. closer to the reading for precision.

COM - on common. Black probe goes here

m A (most commonly used)

10 A (for very large current)

measuring battery voltage

• set knob at 20V

Red ⚡ Black

1.5V

2000 mV (more precise) (Best)

20V → 1.53
2000mV → 1.532
200V → 0.15

We can also use a multimeter to check continuity (if a circuit is complete).
Step Set the knob to diode or resistance and if u get a small resistance the wires are connected, the no. stays same for disconnected wire (0 or 1).

Resistors: Colour bands on resistor helps identify its ohm.

e.g.  → 10 kΩ ±5%

How to identify:

1st digit, 2nd digit, Multiplier, tolerance

$\Omega = (1st\ band) \times (2nd\ band) \times 10^{multiplier}$

check using multimeter:
Set knob to 20 kΩ and touch both ends with knobs (colour side does matter - not polar). Check value.

If you don't know the range of the resistor, set the knob to a high value and the lower it to appropriate no. on 2nd try for more significant figure answer.

code:

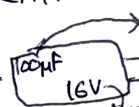
Black	0
Brown	1
Red	2
Orange	3
Yellow	4
Green	5
Blue	6
Violet	7
Gray	8
White	9

Tolerance:

Gold	±5%
Silver	±10%

Capacitors: Capacitors with uneven tail, is polar. The longer leg should be connected to a higher voltage

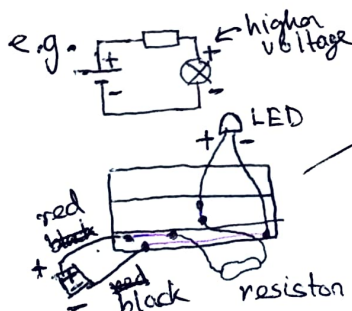
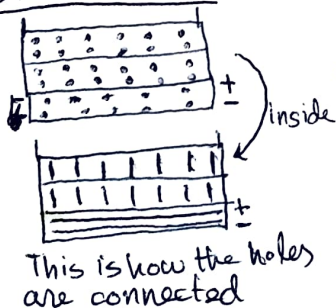
Non polar ceramic capacitors can be connected either way

e.g.  → capacitance

voltage rating - can withstand max 16V

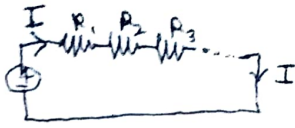
+ → red, black, high voltage

Breadboard:



we can now use the multi meter across the resistor to get the voltage - And the, $I = \frac{V}{R}$ to get current. (Measure resistance beforehand)

9

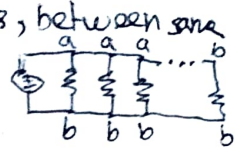
- Series: Same current flows through all component. 

$$R_{total} = R_1 + R_2 + R_3 + \dots$$

Voltage gets divided among components. Calculated using $V = RI$

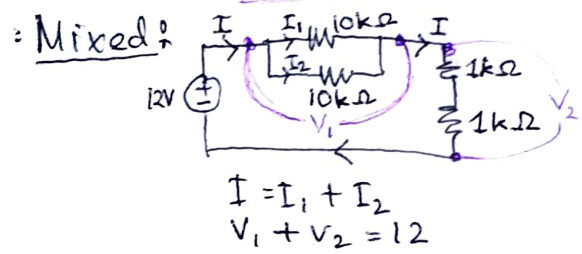
- Parallel: Component connected across each other's leads, between same two points.

$$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$



Voltage same for all components.

Current divided between components. $I = \frac{V}{R}$

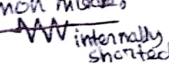


$$I = I_1 + I_2$$

$$V_1 + V_2 = 12$$

Construct the circuit in the breadboard. Measure resistance of the resistors and voltage at V_1 and V_2 using the multimeter. (connect carefully). Calculate to get I, I_1, I_2

- Installing Pspice 3 (file in F:\) Do as it says on the "Read me" txt file. Once done, search "Pspice Design Manager" on start menu. Click "Run schematics" on the left and will get your work area. Click "Get new part" on top middle to open library for components.

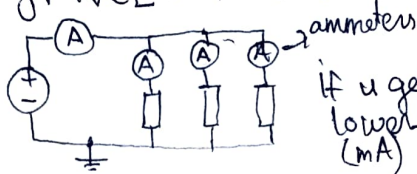
- "Ctrl+G" can select stuff and double click and put as many on the screen. Right click to stop.
- "R" → Resistor  common mistake: internally shorted
- "VDC" → Voltage source (d.c)
- "gnd" (analogue on the other) → ground [important]
- To rotate: click → go to "edit" → click "rotate" or Ctrl+R

- "Draw wires" on top to draw wire
- To set value of components, double click and change it
- "Simulate" (it will show "complete" at the bottom of another window)
 - In this state, if u click "V" at top it'll show the voltage or "I"
 - To get the ΔV on a component, minus the V value of its both side.
 - If you click on the "I" value, it'll show direction.

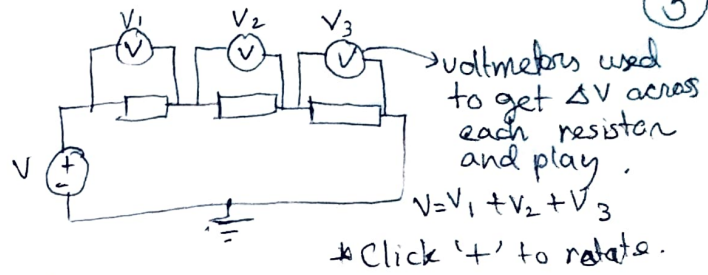
- Proteus: (info in CSE 260) Press P for library, Search "resistor" and click "Generic" is sub categories and double click "RES".
- "Battery" → dc, Ground in "terminal mode" on left
- To rotate: right click on it and "rotate"
- Double click to edit component value.
- Voltmeter on the left in "Instruments" as "DC Voltmeter"
- Attach ammeter in series & voltmeter in parallel and play at bottom to simulate.
- this might show 0. Double click and set unit to μA .
- U can use probes too for V and I at random point in "Probes" on the left.

• Verifying KVL in Proteus:

• Verify KCL in Proteus:

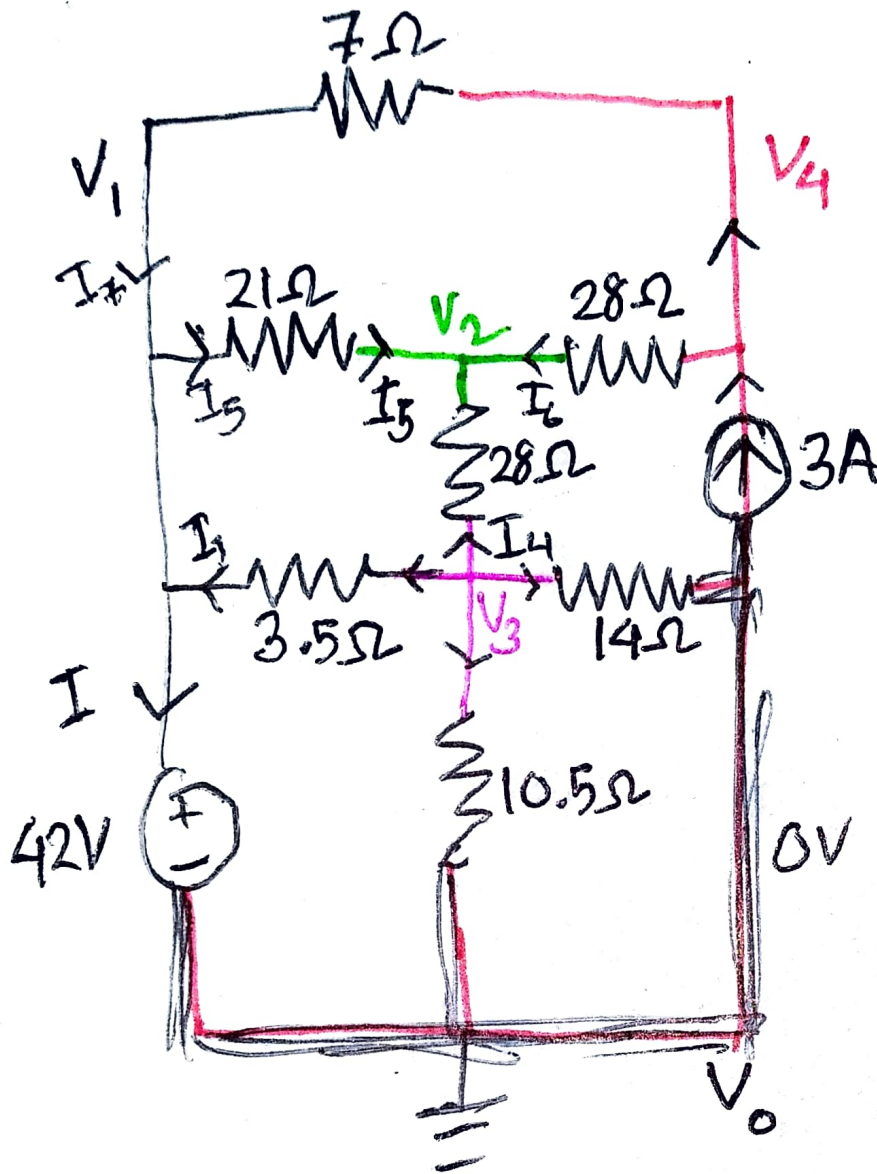


ammeters
if u get 0.00A,
lower the range of the ammeter.
(mA)



• Superposition in Proteus: get the i values by replacing voltage sources with 0V and current sources with 0A which are not part from the source being studied. (Only independent sources). Add a Ammeter to get the values. Add the currents to get expected values using superposition theory.
Restore back the sources and see the i value and compare.

Node Analysis

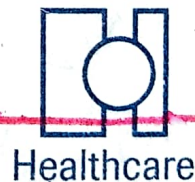


Node 1: $V_1 = 42$ — (i)

Node 2: $I_5 + I_5 + I_6 = 0$

$\Rightarrow \frac{V_2 - V_1}{21} + \frac{V_2 - V_3}{28} + \frac{V_2 - V_4}{28} = 0$ — (ii)

Node 3: $\frac{V_3 - V_1}{3.5} + \frac{V_3 - V_2}{28} + \frac{V_3 - 0}{10.5} + \frac{V_3 - V_4}{14} = 0$ — (iii)



$$\text{Node 4: } \frac{V_4 - V_1}{7} + \frac{V_4 - V_2}{28} - I_B = 0$$

(iv)

$$\text{Solve eq}^n: \frac{V_4 - 42}{7} - \frac{42}{7} + \frac{V_4 - V_2}{28} - 3 = 0 \rightarrow (i)$$

$$\Rightarrow \frac{5}{28} V_4 - \frac{1}{28} V_2 = 9 \quad \text{--- (iv)}$$

$$\frac{5}{42} V_2 - \frac{1}{28} V_3 - \frac{1}{28} V_4 = 2 \quad \text{--- (ii)}$$

$$\frac{41}{84} V_3 - \frac{1}{28} V_2 = 12 \quad \text{--- (iii)}$$

Put these 3 eqⁿ directly in calculator for 3 variable simultaneous equations.

$$\frac{5}{42} V_2 - \frac{1}{28} V_3 - \frac{1}{28} V_4 = 2$$

$$-\frac{1}{28} V_2 + \frac{5}{28} V_4 = 9$$

$$-\frac{1}{28} V_2 + \frac{41}{84} V_3 = 12$$

$$V_2 = 42.80, V_3 = 27.72, V_4 = 58.96$$

• Current through voltage source: $I = I_1 + I_2 - I_3$

$$= \frac{V_3 - V_1}{3.5} + \frac{V_4 - V_1}{7} - \frac{V_1 - V_2}{21}$$

Power through battery = $IV = -68W$

$$= -1.619 A$$